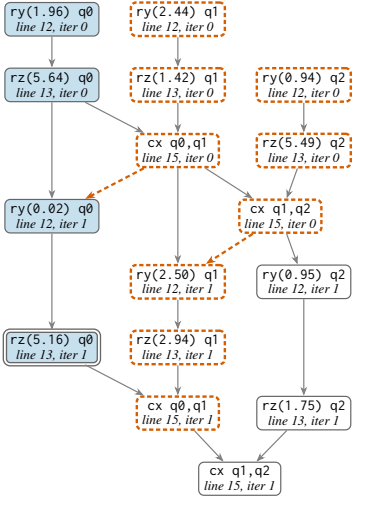
(a) VQE ansatz

```
4  n = 3
5  layers = 2
6  rng = np.random.default_rng(7)
7
8  qc = QuantumCircuit(n, n)
9
10 for layer in range(layers):
11     for qubit in range(n):
12         qc.ry(float(rng.uniform(0, np.pi)), qubit)
13         qc.rz(float(rng.uniform(0, 2 * np.pi)), qubit)
14     for qubit in range(n - 1):
15         qc.cx(qubit, qubit + 1)
```

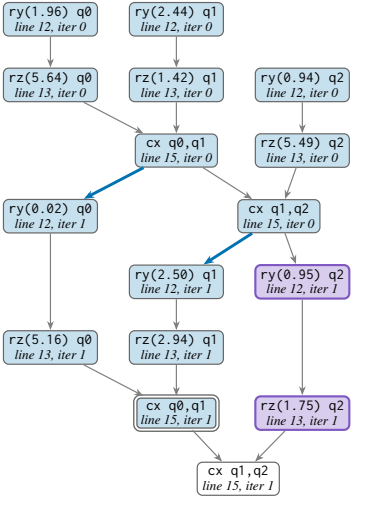
data
quantum
quantum edge missing from the classical PDG
in the slice
dependence the slice misses
added by the QPDG, not a dependence
last write to the criterion

(b) Classical PDG



slice of q_0: 4 of 16 operations
fidelity 0.77, misses 9 dependences

(c) QDG



slice of q_0: 13 of 16 operations
fidelity 1.00, exactly the dependences